ProductItem
-String productId
-String title
-BigDecimal price
-int totalStock
-int version
+hasStock(int qty) : bool
«enumeration»
ReservationStatus
PENDING
CONFIRMED
EXPIRED
RELEASED
InventoryReservation
-String reservationId
-String userId
-String productId
-int quantity
-Instant expiresAt
-ReservationStatus status
+isExpired() : bool
+confirm() : void
+release() : void
Has State
1
References
1
Order
-String orderId
-String userId
-String reservationId
-BigDecimal totalAmount
-String idempotencyKey
-OrderStatus status
+markPaid() : void
+cancel() : void
Has State
«enumeration»
OrderStatus
CREATED
PAYMENT_PROCESSING
PAID
CANCELLED
«interface»
IOrderRepository
+save(Order order) : Order
+findByIdempotencyKey(String key) : Order
Uses
«interface»
IInventoryRepository
+reserveAtomicLua(String productId, int qty, long ttlSeconds) : bool
+releaseStock(String productId, int qty) : void
+saveReservation(InventoryReservation res) : void
+findByReservationId(String resId) : InventoryReservation
Uses
Implements (Repository Pattern)
RedisPostgresInventoryRepository
-RedisTemplate redisTemplate
-PostgresConnection pgConn
+reserveAtomicLua(String productId, int qty, long ttlSeconds) : bool
+releaseStock(String productId, int qty) : void
+saveReservation(InventoryReservation res) : void
+findByReservationId(String resId) : InventoryReservation
CheckoutOrderService
-IOrderRepository orderRepo
-ReservationService reservationService
-IPaymentAdapter paymentAdapter
-KafkaDomainEventPublisher eventPublisher
+placeOrder(String userId, String resId, String idemKey) : Order
Delegates Charge
«interface»
IPaymentAdapter
+processPayment(String orderId, BigDecimal amount, String idemKey) : PaymentResult
Implements (Adapter Pattern)
StripePaymentAdapter
-StripeClient stripeClient
+processPayment(String orderId, BigDecimal amount, String idemKey) : PaymentResult
Validates TTL
ReservationService
-IInventoryRepository inventoryRepo
-KafkaDomainEventPublisher eventPublisher
+createReservation(String userId, String productId, int qty) : InventoryReservation
+handleTimeout(String reservationId) : void
Publishes Events
KafkaDomainEventPublisher
-KafkaTemplate kafkaTemplate
-List<IEventObserver> observers
+registerObserver(IEventObserver observer) : void
+notifyObservers(DomainEvent event) : void
«interface»
IEventObserver
+onEvent(DomainEvent event) : void
Implements (Observer Pattern)
InventoryRollbackObserver
-ReservationService reservationService
+onEvent(DomainEvent event) : void
Triggers Rollback
Emits OrderEvents
Notifies (Observer Pattern)